THE TOP CHERN CLASS OF A FINITE LOCALLY FREE MODULE OF RANK R CAPS THE FUNDAMENTAL CLASS TO THE CYCLE OF A REGULAR SECTION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The Chern class c_r(E) is a bivariant class. Same chapter chow.tex, label lemma-cap-cp-bivariant, invoked at proof line 8765. Allows the base change X' -> X to be absorbed, reducing the statement to the case X = X'; the rest of the argument stays in the counted span.
- The geometric case of the top Chern class for a regular section. Same chapter chow.tex, label lemma-geometric-cap, invoked at proof lines 8770 and 8820. Supplies the case r = 1 and is applied again to prove the auxiliary claim [Z(t)]_{n+r-2} = c_1(O_P(1)) cap [P]_{n+r-1}.
- The projective space bundle formula and depth along a flat morphism. Same chapter chow.tex, label lemma-chow-ring-projective-bundle, invoked at proof line 8779; Algebra, labels algebra-lemma-flat-increases-depth (proof line 8785) and algebra-lemma-grothendieck (proof line 8831). Lets the equality be checked after pulling back along the projective bundle of E, keeps the regular-sequence hypothesis under flat pullback, and supplies the local nonzerodivisor criterion used in the claim.

Source reference. Original lemma 00LM: sources/stacks-chow/tags.txt, source lines 792–792. Navigation pointer for \ref{algebra-lemma-flat-increases-depth} used inside the counted span (line 8785); the argument that consumes it is retained in the span.

Source reference. Original lemma 00MF: sources/stacks-chow/tags.txt, source lines 821–821. Navigation pointer for \ref{algebra-lemma-grothendieck} used inside the counted span (line 8831); the argument that consumes it is retained in the span.

Date: stacks-project:chow.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0B7H: sources/stacks-chow/pinned-original-source.tex, source lines 7533–7533. Navigation pointer for \ref{lemma-cap-cp-bivariant} used inside the counted span (line 8765); the argument that consumes it is retained in the span.

Source reference. Original lemma 02TX: sources/stacks-chow/pinned-original-source.tex, source lines 6932–6932. Navigation pointer for \ref{lemma-chow-ring-projective-bundle} used inside the counted span (line 8779); the argument that consumes it is retained in the span.

Source reference. Original lemma 02SQ: sources/stacks-chow/pinned-original-source.tex, source lines 4432–4432. Navigation pointer for \ref{lemma-geometric-cap} used inside the counted span (line 8770); the argument that consumes it is retained in the span.

Lemma 0.1. *In the situation described just above assume* $\dim_\delta(X') = n$*, that* $f^*\mathcal{E}$ *has constant rank* r*, that* $\dim_\delta(Z(s)) \leq n - r$*, and that for every generic point* $\xi \in Z(s)$ *with* $\delta(\xi) = n - r$ *the ideal of* $Z(s)$ *in* $\mathcal{O}_{X',\xi}$ *is generated by a regular sequence of length* r*. Then*

$$c_r(\mathcal{E}) \cap [X']_n = [Z(s)]_{n-r}$$

in $\mathrm{CH}_*(X')$*.*

Proof. Since $c_r(\mathcal{E})$ is a bivariant class (Lemma 0B7H) we may assume $X = X'$ and we have to show that $c_r(\mathcal{E}) \cap [X]_n = [Z(s)]_{n-r}$ in $\mathrm{CH}_{n-r}(X)$. We will prove the lemma by induction on $r \geq 0$. (The case $r = 0$ is trivial.) The case $r = 1$ is handled by Lemma 02SQ. Assume $r > 1$.

Let $\pi : P \to X$ be the projective space bundle associated to $\mathcal{E}$ and consider the short exact sequence

$$0 \to \mathcal{E}' \to \pi^*\mathcal{E} \to \mathcal{O}_P(1) \to 0$$

By the projective space bundle formula (Lemma 02TX) it suffices to prove the equality after pulling back by π. Observe that $\pi^{-1}Z(s) = Z(\pi^*s)$ has δ-dimension $\leq n-1$ and that the assumption on regular sequences at generic points of δ-dimension $n-1$ holds by flat pullback, see Algebra, Lemma 00LM. Let $t \in \Gamma(P, \mathcal{O}_P(1))$ be the image of π^*s. We claim

$$[Z(t)]_{n+r-2} = c_1(\mathcal{O}_P(1)) \cap [P]_{n+r-1}$$

Assuming the claim we finish the proof as follows. The restriction $\pi^*s|_{Z(t)}$ maps to zero in $\mathcal{O}_P(1)|_{Z(t)}$ hence comes from a unique element $s' \in \Gamma(Z(t), \mathcal{E}'|_{Z(t)})$. Note that $Z(s') = Z(\pi^*s)$ as closed subschemes of P. If $\xi \in Z(s')$ is a generic point with $\delta(\xi) = n - 1$, then the ideal of $Z(s')$ in $\mathcal{O}_{Z(t),\xi}$ can be generated by a regular sequence of length $r - 1$: it is generated by $r - 1$ elements which are the images of $r - 1$ elements in $\mathcal{O}_{P,\xi}$ which together with a generator of the ideal of $Z(t)$ in $\mathcal{O}_{P,\xi}$ form a regular sequence of length r in $\mathcal{O}_{P,\xi}$. Hence we can apply the induction hypothesis to s' on $Z(t)$ to get $c_{r-1}(\mathcal{E}') \cap [Z(t)]_{n+r-2} = [Z(s')]_{n-1}$. Combining all of the above we obtain

$$\begin{aligned} c_r(\pi^*\mathcal{E}) \cap [P]_{n+r-1} &= c_{r-1}(\mathcal{E}') \cap c_1(\mathcal{O}_P(1)) \cap [P]_{n+r-1} \\ &= c_{r-1}(\mathcal{E}') \cap [Z(t)]_{n+r-2} \\ &= [Z(s')]_{n-1} \\ &= [Z(\pi^*s)]_{n-1} \end{aligned}$$

which is what we had to show.

Proof of the claim. This will follow from an application of the already used Lemma 02SQ. We have $\pi^{-1}(Z(s)) = Z(\pi^* s) \subset Z(t)$. On the other hand, for $x \in X$ if $P_x \subset Z(t)$, then $t|_{P_x} = 0$ which implies that s is zero in the fibre $\mathcal{E} \otimes \kappa(x)$, which implies $x \in Z(s)$. It follows that $\dim_\delta(Z(t)) \leq n + (r-1) - 1$. Finally, let $\xi \in Z(t)$ be a generic point with $\delta(\xi) = n + r - 2$. If ξ is not the generic point of the fibre of $P \to X$ it is immediate that a local equation of $Z(t)$ is a nonzerodivisor in $\mathcal{O}_{P,\xi}$ (because we can check this on the fibre by Algebra, Lemma 00MF). If ξ is the generic point of a fibre, then $x = \pi(\xi) \in Z(s)$ and $\delta(x) = n + r - 2 - (r-1) = n - 1$. This is a contradiction with $\dim_\delta(Z(s)) \leq n - r$ because $r > 1$ so this case doesn't happen. $\square$